

Charge & Field

ChatGPT edition / Investigation 1

Map an invisible interaction without pretending the arrows are objects

Question

How can the direction and relative strength of an electric interaction be recorded when we cannot see the field itself?

Build a repeatable detector

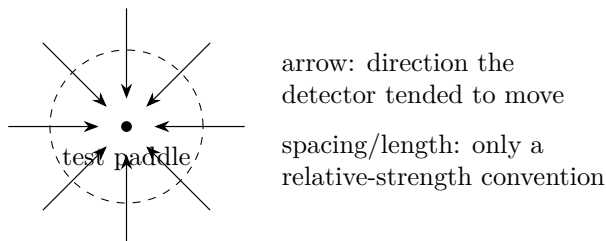
Materials: clear tape, two drinking straws, thread, index card, wool or clean dry hair. No outlet-powered device is used.

1. Hang a 12 cm tape strip from thread as a light “paddle”; fold the last centimetre into a handle. Make a second identical strip.
2. Press both strips to the same clean tabletop, label the handles A and B, then pull them up by the handles. Bring B toward hanging A without touching.
3. Make a third strip C. Press A and C together face-to-back, peel them apart, and compare A–C with A–B. Record attraction or repulsion, distance, and motion.
4. Repeat three times. Humidity can weaken the effect; “no motion” is data.

KEEP SCALE HONEST

This static-tape investigation is not permission to experiment with outlets, CRT displays, ignition coils, stun devices, Van de Graaff terminals, or storms. Those sources differ radically in available energy and voltage. **STOP: no outlets, mains leads, wall adapters, large battery packs, large capacitors, ignition coils, or exposed high voltage.**

Map an effect, not an imaginary substance



On an index card, mark detector positions 1, 2, 4 and 8 cm from a charged strip. At each position, release the detector from the same angle and rate the initial motion 0–3. Repeat, then draw arrows. State what your map directly records and what it infers.

Evidence record

Prediction	_____
Changed variable	_____
Measured result	_____
Claim + because	_____

The charge scale

Charge comes in opposite signs; like signs repel and unlike signs attract. Macroscopic charge is assembled from elementary charges whose magnitude is

$$e = 1.602\,176\,634 \times 10^{-19} \text{ C} \quad (\text{exact in SI}).$$

The experiment identifies relative sign; it does not tell whether the moving charge carriers were positive or negative.

Extension: test a model
Predict what doubling distance will do to your motion rating. Make five trials at each distance. Does the evidence support “same,” “steadily weaker,” or an exact inverse-square law? Explain why a tape paddle is too crude for the last claim.

Evidence: NIST S05; DOE S06. Safety boundary: OSHA S01–S03. See the provider-neutral electricity source register.

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